

✓ Homework 7: Marketing

After going through the lecture notebooks, work on the homework below.

Question

Using the MyAnimeList dataset [1], perform **Decile Analysis** on anime popularity.

1. Divide anime into **10 equal-frequency groups (deciles)** based on `Members` using `pd.qcut`, with labels `['01', '02', '03', '04', '05', '06', '07', '08', '09', '10']` (where `'01'` is the lowest `Members` group).
2. For each decile group, calculate:
 - **MeanScore**: Mean of `Score`
 - **TotalMembers**: Sum of `Members`
3. Calculate **CumulativePct**: Cumulative percentage of `TotalMembers` relative to the overall total (as a value from 0 to 100).
4. Return a `pd.DataFrame` indexed by the decile group label (`'01'` to `'10'`) with columns `['MeanScore', 'TotalMembers', 'CumulativePct']`.

Inputs / Outputs:

```
homework(anime_data_extracted: pd.DataFrame) -> pd.DataFrame
```

Argument	Type	Description
<code>anime_data_extracted</code>	<code>pd.DataFrame</code>	Preprocessed anime dataset (see Section 2)
Return	Type	Description
<code>result</code>	<code>pd.DataFrame</code>	DataFrame indexed by decile label with columns <code>['MeanScore', 'TotalMembers', 'CumulativePct']</code>

Example:

```
result = homework(anime_data_extracted)
print(result)
```

```

      MeanScore  TotalMembers  CumulativePct
Members
```

01	5.5XXXXX	4XXXXX	0.08XXXX
02	5.7XXXXX	10XXXXX	0.25XXXX
03	5.9XXXXX	18XXXXX	0.56XXXX
04	6.1XXXXX	31XXXXX	1.08XXXX
05	6.4XXXXX	51XXXXX	1.94XXXX
06	6.6XXXXX	87XXXXX	3.38XXXX
07	6.7XXXXX	164XXXXX	6.11XXXX
08	6.9XXXXX	346XXXXX	11.86XXXX
09	7.1XXXXX	878XXXXX	26.43XXXX
10	7.6XXXXX	4434XXXXX	100.000000

Submission Guidelines:

1. Submit the entire `homework()` function by selecting this week's assignment in the Omnicampus homework section, pasting the function into the submission area, and then clicking [Submit Python Code].

Note:

- Erase the `!!WRITE ME!!` when submitting
- The return value of the function should be type `pd.DataFrame`
- The DataFrame should be indexed by the decile group label and have columns `['MeanScore', 'TotalMembers', 'CumulativePct']` in this order
- Write your answer as one function
- You may use any Generative AI (NotebookLM recommended)

Deadline

Wed, Jun 17 (11:00 UTC)

✓ 1. Importing Libraries

Note: Do not include this section in your `homework()` function.

```
import numpy as np
import pandas as pd
from pandas import DataFrame

# Libraries for retrieving data from the web and handling zip files
import requests, zipfile
from io import StringIO
import io
```

✓ 2. Downloading the Dataset

Download the MyAnimeList dataset from GitHub and clean the data by removing rows where `Score` is `'Unknown'`.

Note: Do not include this section in your `homework()` function.

```
# Specify the url with data
url = 'https://github.com/Hernan4444/MyAnimeList-Database/archive/refs/heads/master.zip'

# Acquire data from the url
r = requests.get(url, stream=True)

# read and extract the zipfile
z = zipfile.ZipFile(io.BytesIO(r.content))
z.extractall()

# Load and clean the data
anime_data = pd.read_csv('MyAnimeList-Database-master/data/anime.csv')
anime_data_extracted = anime_data[anime_data['Score'] != 'Unknown'].copy()
anime_data_extracted['Score'] = pd.to_numeric(anime_data_extracted['Score'])
```

✓ 3. Solution

Write your solution in the `homework()` function below. The function receives `anime_data_extracted` as input and should return a `pd.DataFrame`.

```
# Fill in the WRITE ME
# Don't forget to erase the "!!WRITE ME!!" before submitting!
def homework(anime_data_extracted):
    !!WRITE ME!!
```

✓ 4. Test Your Solution

Run the test cell below to check that your `homework()` function does not contain import statements or data loading code before submitting.

Note: Do not modify the test code.

```
import ast
import inspect
import textwrap
```

```

def hw7_public_tests(homework_func):
    print("=== HW7 Public Tests ===")

    # Get the source code of the homework function
    source = textwrap.dedent(inspect.getsource(homework_func))
    tree = ast.parse(source)

    # ----- Test 1: No import statements -----
    print("Test 1: No import statements in function...", end=" ")
    for node in ast.walk(tree):
        if isinstance(node, (ast.Import, ast.ImportFrom)):
            print("NG")
            if isinstance(node, ast.Import):
                names = ", ".join(alias.name for alias in node.names)
            else:
                names = node.module
            print(f" Found import statement: '{names}'")
            print(" Remove all import statements before submitting.")
            return
    print("OK")

    # ----- Test 2: No data loading code -----
    print("Test 2: No data loading code in function...", end=" ")
    blocked_calls = {"read_csv", "read_excel", "read_json", "read_html",
                     "get", "post", "ZipFile", "extractall", "urlopen"}
    for node in ast.walk(tree):
        if isinstance(node, ast.Call):
            func_name = ""
            if isinstance(node.func, ast.Attribute):
                func_name = node.func.attr
            elif isinstance(node.func, ast.Name):
                func_name = node.func.id
            if func_name in blocked_calls:
                print("NG")
                print(f" Found data loading call: '{func_name}()'")
                print(" Do not include data downloading/loading code in your f")
                return
    print("OK")

    # ----- Test 3: No hardcoded file paths or URLs -----
    print("Test 3: No hardcoded file paths or URLs...", end=" ")
    for node in ast.walk(tree):
        if isinstance(node, ast.Constant) and isinstance(node.value, str):
            val = node.value
            if val.startswith("http") or ".csv" in val or ".zip" in val or ".xl
            print("NG")
            print(f" Found hardcoded path/URL: '{val[:60]}...'")
            print(" Do not include file paths or URLs in your function.")
            return
    print("OK")

```

```
print("=== All Public Tests Passed ===")  
  
hw7_public_tests(homework)
```

Once all tests have passed, copy and paste your `homework()` function to Omnicampus and submit your solution. If you see a `3.0`, it means that your answer was correct.

✓ References

[1] Hernan4444. *MyAnimeList Database*. GitHub.

<https://github.com/Hernan4444/MyAnimeList-Database>

Double-click (or enter) to edit